JieZi: A Large-Scale Expert-Audited Dataset and Benchmark for Ancient Chinese Character Exegesis Supplementary Material

This supplementary material provides additional details that could not be included in the main paper due to space constraints. First, Sec. A provides scholarly background on the ACCE task formulation and illustrative examples. Second, Sec. B presents further details on dataset construction, including data distribution, prompt templates, and quantitative verification statistics. Finally, Sec. C provides additional experimental details and the complete results on JieZi-Bench.

A Task Formulation Details

A.1 Scholarly Background of ACCE

Ancient character exegesis. In Chinese paleography, *exegesis* refers to the scholarly practice of interpreting an ancient glyph by jointly analyzing its visual form, internal structure, semantic content, and historical evolution [12, 25, 26]. As one of the oldest continuously used writing systems, ancient Chinese characters carry irreplaceable value for research in archaeology, history, and historical linguistics [8, 11], making their systematic interpretation a long-standing scholarly priority. Unlike modern character recognition, which maps a glyph to a single Unicode label, exegesis requires the scholar to explain *why* a character has its particular form, meaning, and evolutionary trajectory. In practice, scholars follow a progressive analytical workflow [2, 20] that can be decomposed into four sequential stages: **Step 1: Identification.** The scholar begins by **identifying the character** and **determining which historical script period** it belongs to. **Step 2: Structural Analysis.** The scholar then analyzes the glyph's **internal structure** by **decomposing it into components** and examining how each component contributes to the character's form and meaning. **Step 3: Semantic Interpretation.** Building on this structural understanding, the scholar interprets the character's **original semantic value** in its historical context. **Step 4: Diachronic Tracing.** Finally, the scholar traces how both **form and meaning evolved** across successive script periods, explaining the motivations behind these diachronic changes. This multi-stage, knowledge-intensive process constitutes the standard methodology taught in Chinese philology programs and practiced in archaeological and linguistic research.

Challenges in exegesis. Exegesis is particularly challenging because it demands the simultaneous integration of heterogeneous knowledge types: visual pattern recognition for identifying glyph components, linguistic knowledge for understanding component functions, semantic reasoning for inferring original meaning, and historical knowledge for explaining diachronic change. Each requires different cognitive skills and a distinct body of expertise. Moreover, ancient glyphs exhibit high visual variability; the same character may appear radically different across script periods (e.g., OBI, Bronze, Seal, Clerical), and different characters may share visually similar components, making identification and structural analysis error-prone even for trained scholars. The reasoning involved is also inherently layered: errors in early stages (e.g., misidentifying a component) cascade into incorrect semantic and evolutionary interpretations, meaning that partial understanding is often insufficient and the task cannot be reduced to independent classification problems. As a result, training a human expert in character exegesis typically requires years of graduate-level study, and the analysis of a single character may involve cross-referencing multiple authoritative dictionaries and reconciling conflicting scholarly interpretations [6, 19].

Formalization as a computational task. Despite its importance, exegesis has received limited attention from the computational community. Existing work on ancient Chinese characters focuses predominantly on recognition and retrieval [7, 13], which correspond to only the most elementary step of the full exegetical workflow. To our knowledge, no prior work has attempted to formalize the complete, multi-stage scholarly process as a computational task. ACCE addresses this gap by decomposing exegesis into four progressive levels that directly mirror the stages of the scholarly workflow described above: identifying a character and its script period (**L1: Basic Information**), analyzing internal structure and component functions (**L2: Glyph Form**), interpreting the original semantic value (**L3: Glyph Meaning**), and tracing diachronic evolution (**L4: Diachronic Evolution**). This decomposition preserves the layered and dependent nature of the original scholarly practice while making each stage independently measurable. It enables researchers to construct targeted supervision, apply stage-specific evaluation metrics, and identify precisely which aspects of exegesis current models can and cannot handle.

A.2 Examples of Tasks

To provide a more intuitive understanding of the ACCE task, we present qualitative examples from JieZi-Dataset. Fig. 2 illustrates the ten fine-grained subtasks. While these subtasks assess individual capabilities, the ultimate goal of ACCE is holistic interpretation. Fig. 1 presents comprehensive scholarly exegesis in real scenarios, where models are challenged with complex paleographic phenomena such as severe structural corruption and visually similar but semantically distinct glyph forms. As demonstrated, a high-quality exegesis successfully resolves these challenges by synthesizing all four progressive levels (L1 to L4) into a coherent, domain-aligned explanation. Fig. 3 contrasts the responses of the fine-tuned model with those of baseline models on representative subtasks.

Question: Give a scholarly exegesis for the Ancient Chinese Character. Image :

Answer: This is the seal script form of the Chinese character **bèi (背)**. A two-component **Phonogram character (形声字)** , it combines ***běi (北)*** (phonetic-semantic, depicting two people back-to-back) and **(肉)** (semantic radical for body parts, later standardized as **(月) ròu**. Its original meaning is **"the human back/spine"**. Derived from 北, it evolved into the regular script 背 after clerical transformation.

- L1: Basic
- L2: Form
- L3: Meaning
- L4: Evolution

Question: The left component appears to be "body" (身). What is the meaning of this component? Give a scholarly exegesis revealing its true origins. Image :

Answer: This is the seal script form of the character **shè (射)**. It is an **Ideogram (会意字)**. While it visually appears to consist of **(身) "body"** on the left and **(寸) "hand"** on the right, its true paleographic components are actually a bow, an arrow, and a hand. Its original meaning is **"to draw a bow and shoot an arrow"**. In earlier scripts, it depicted a hand (又) releasing an arrow from a bow. However, during the seal script stage, **the "bow" shape was severely corrupted (讹变) into "body" (身)**. This visual corruption caused later scholars (in *Shuowen Jiezi*) to completely misinterpret its etymology.

- L1: Basic
- L2: Form
- L3: Meaning
- L4: Evolution

Question: This ancient glyph shares almost no visual traits with its modern character **wǎng** (往). Give a scholarly exegesis explaining its evolution. Image :

Answer: This is OBI of **wǎng** (往). It is a phono-semantic compound (形声兼会意). The glyph purely consists of **(止) "foot"** indicating movement, and **(王) "king"** acting as the phonetic component. Its original meaning is **"to go"**. It underwent severe structural corruption (讹变). Later, the semantic radical **(彳) "step"** was added to the left, while the original "止+王" visually mutated into the completely unrecognizable component **(主)**, forming the modern regular script **往**.

- L1: Basic
- L2: Form
- L3: Meaning
- L4: Evolution

Figure 1: Examples of comprehensive scholarly exegesis in complex paleographic scenarios.

B Dataset Construction and Quality Control

B.1 Data Distribution and Metrics

JieZi-Dataset. To illustrate the data diversity and comprehensiveness, Tab. 1 presents the subtask coverage distribution within the JieZi-Dataset. Unlike the individually targeted test queries in JieZi-Bench, the training data often features comprehensive QAs that simultaneously address multiple analytical dimensions, mirroring the holistic nature of real-world scholarly exegesis. Consequently, the sum of subtask frequencies exceeds the total number of unique QA pairs.

JieZi-Bench. Tab. 2 summarizes the question distribution and corresponding evaluation metrics across the ACCE tasks in JieZi-Bench. Notably, since component-related subtasks (COMR, COMF, COMI, and COME) are evaluated collectively within unified QAs, they are grouped as Component Analysis. Within this group, each subtask uses a distinct evaluation metric.

B.2 Prompt for Structurization

Fig. 4 and Fig. 5 present the prompts used during Structurization for extracting structured metadata from image-text pairs.

B.3 Templates for VQA generation

Figure 6 presents the templates used during generating QA pairs.

COMR

Question: What character is this glyph? Image :

Answer: This is character **sī (司)**.

- L1: Basic

SCRC

Question: What is the script type of this glyph? (OBI, Bronze, Warring state, Seal, Clerical, Regular, Cursive) Image :

Answer: This is a **Bronze** script.

- L1: Basic

STRC

Question: What is the structure of this glyph? Image :

Answer: **Left-right structure.**

- L2: Form

COMR

Question: What is the component on the left? Image :

Answer: 言(yán)

- L2: Form

COMF

Question: What is the function of the component '贝'? (Semantic, Phonetic, Symbol) Image :

Answer: Semantic

- L2: Form

COMI

Question: What is the interpretation of the component '贝'? Image :

Answer: It resembles the form of a shell currency, signifying valuables.

- L2: Form

FORC

Question: Which of the Six Scripts does this glyph belong to?(Pictogram, Ideograms, Ideogrammic compounds, Semantic-phonetic, Derivative, Borrow) Image :

Answer: Pictogram

- L2: Form

ORIM

Question: What is the original meaning of this glyph? Image :

Answer: It originally refers to **the seasons of the year**.

- L3: Meaning

COME

Question: What is the type of evolution for the component '厂' in this glyph? (Retention, Corruption, Simplification, Elaboration) Image :

Answer: Corruption

- L4: Evolution

EVOI

Question: What is the evolution of the character in this glyph? Image :

Answer: Historically, it evolved from an OBI depicting a child in the belly, to a Bronze Script showing a baby gestating in a placenta, and finally to a Seal Script formed by '勹' (wrapping) and '巳' (fetus). The modern standard character presents a semi-enclosed structure, where the outer radical wraps around the inner component.

- L4: Evolution

Figure 2: Examples of Question-Answering pairs across the ten fine-grained sub-tasks within the ACCE framework.

Question: Give a scholarly exegesis for the Ancient Chinese Character.

Image :

Doubao-seed-2.0-pro:
This character consists of three retained semantic components: the '宀'-shaped outer frame representing the granary building, the measuring vessel reflecting its storage function, and the grain (粟) denoting the stored food.

Qwen3.5-9B + JieZi-Dataset
This character is composed of three retained components: the semantic '宀' (representing a house or covering) and semantic '爿' (representing a bed), plus the phonetic '未' which provides the sound.

Question: Explain the historical evolution of the glyph in the image.

Image :

Doubao-seed-2.0-pro:
This character is a phono-semantic compound with '心' as the semantic component and '满' as the phonetic component. As it developed into the Regular Script, it retained its top-bottom structure. Its core components have not undergone significant graphical corruption, keeping the overall form regular and stable.

Qwen3.5-9B + JieZi-Dataset
The character was originally written as '懣' (composed of the semantic '心' and phonetic '菌'), with its original meaning being 'vexed' or 'depressed'. A later variant emerged using '满' as the phonetic component. In the modern simplified character, the phonetic part '满' is simplified while the semantic part '心' is retained, finalizing its standard form as '懑'.

Figure 3: Examples of responses for different tasks from different models. Red text indicates error messages, and blue text indicates correct messages.

B.4 Quantitative Verification Details

To make the quality-control process more transparent, we report the verification scope, sampling strategy, and revision statistics for each stage of our pipeline.

Stage 1: Collection. For JieZi-Dataset, Stage 1 mainly involves digitization, including OCR correction and alignment of the digitized content with the original dictionary pages. Since the goal at this stage is only to ensure fidelity to the source dictionary rather than paleographic interpretation, the full scanned volume (over 2,000 pages) was checked by general annotators without requiring domain expertise. The main error types were OCR omissions and misrecognition of variant character forms. While errors were found in less than 1% of entries, this full manual verification remains indispensable: as noted in the main paper, OCR errors disproportionately concentrate on rare and archaic glyphs, which are precisely the characters most critical to ACCE. Without exhaustive correction, even a small number of such errors would propagate into downstream structuring and QA generation, silently degrading the most informative entries in the dataset.

<Template> Extraction

Role
You are a top-level classical Chinese dictionary data-cleaning expert and structured-data engineer. Your task is to convert noisy, inconsistently formatted semi-structured classical Chinese dictionary text into extremely clean JSON that strictly follows the required specification.

Task
Read the single-character dictionary text from the user. Parse fields such as [Original Character], [Pinyin], [Derived/Variant Characters], [Structure], [Original Meaning], and [Evolution], and output strictly according to the required JSON schema.

1) Global cleanup and noise removal
- Remove meaningless layout symbols, especially stray backslashes mixed into Hanzi (for example, clean `\傍` into `傍`).
- Ignore and drop the entire [Glyph Form] section (from [Glyph Form] to the next section header).
- Ignore all non-schema sections (for example [Illustration], [Character Composition], etc.). Do not mix them into `form_chunk` or `meaning_chunks`.
- Do not output image paths, `[image: ...]`, or anything like `extracted_images/...`.
- Keep example sentences as-is (including parenthetical notes). Only do necessary formatting cleanup.

2) Field mapping and merge rules
- `char`: extract from [Original Character].
- `pinyins`:
  - Extract all pinyin from [Pinyin]. Split multiple values by semicolon/comma/space into an array.
  - Important completion: if [Evolution] contains patterns like "通 'X(pinyin)' / 又通 'X(pinyin)'", add that pinyin to global `pinyins` (deduplicated).
- `derived_chars`:
  - Mainly from [Derived Characters]/[Variant Characters]. If missing, output `[]`.
  - Split by separators such as `、`, `;`, `,`, `，`, `；`, or spaces.
  - Optional completion: if [Structure] explicitly says "variant written as X / colloquial form X / also written as X", and X is not already listed, add it (deduplicated).
- `form_chunk`:
  - Keep only cleaned original text from [Structure], and keep the `【构造】` tag at the beginning.
  - Do not merge [Original Meaning] text into `form_chunk`.

3) Meaning splitting (most critical)
- [Evolution] contains all senses. Split thoroughly: each numbered sense in the source text (`1.`, `2.`, `3.`...) must become one object in `meaning_chunks`.
- Use an internal state machine:
  - `pinyin` inheritance: when markers like `㊀` / `㊁` with pinyin appear, update current pinyin; following senses inherit it unless a new one appears.
  - If no pinyin marker appears in [Evolution], default all senses to the first item of global `pinyins`.
  - `pos` inheritance: when explicit POS appears (verb/noun/adjective/onomatopoeia etc.), update current POS; following senses inherit it unless changed.
- `meaning` extraction:
  - Remove numbering and leading filler words (like "originally means / refers to / also refers to / describes"). Keep core meaning.
  - If explicit derivation exists, you may append a parenthetical note to meaning.
- `notes` extraction:
  - Put special usage into `notes` (joined by `；`): original meaning, used as "X", interchangeable with "X", old interchangeable form, also interchangeable with "X(pinyin)", also used as surname, etc.
  - Do not move parenthetical text from examples into notes.
- `examples` processing:
  - Content after colon is usually examples. Original examples may be separated by `|`; normalize to `example1 | example2 | example3`.
  - Clean backslashes in examples only; do not alter other wording.
  - If no examples, set `examples` to `""`.
- Non-empty fallback:
  - `pinyin` and `pos` must never be empty. If missing in source, fill `pinyin` with inherited/current (or first global pinyin), and fill `pos` with `other`.

Figure 4: Prompt of metadata construction.

Table 1: Coverage of ACCE subtasks within the JieZi-Dataset. Since a single comprehensive QA pair may simultaneously address multiple subtasks, the cumulative count across all subtasks exceeds the total number of unique QA pairs (500K).

Level	Subtask	QAs Covering Subtask
L1: Basic	CHAR	175,875
	SCRC	175,875
L2: Form	STRC	175,875
	COMR	175,875
	COMF	221,558
	COMI	221,558
	FORC	312,984
L3: Meaning	ORIM	175,875
L4: Evolution	COME	221,558
	EVOI	210,360
Overall	**Unique QA Pairs**	505,867

Table 2: Distribution of Each Task in JieZi-Bench

Task	Metric	Question Count
CHAR	Accuracy	1024
SCRC	Accuracy	1024
STRC	Accuracy	1024
FORC	Accuracy	1024
ORIM	BERTScore	1024
EVOI	LLM-as-a-Judge	1024
COMR	F1-Score + Accuracy	1852
COMF	F1-Score + Accuracy	1852
COMI	F1-Score + BERTScore	1852
COME	F1-Score + Accuracy	1852
overall	**Unique QA Pairs**	7996

For JieZi-Bench, the source entries were also fully checked at the collection stage. Errors requiring correction were found in approximately 10% of entries, a notably higher rate than that of JieZi-Dataset. This is expected for two reasons: (1) JieZi-Bench integrates entries from multiple authoritative dictionaries, and cross-source alignment and deduplication introduce additional inconsistencies absent from the single-source Dataset; (2) as a benchmark intended for evaluation, JieZi-Bench applies stricter acceptance criteria for entry completeness and accuracy, causing more entries to be flagged for revision. All identified errors were corrected before proceeding to the next stage.

Stage 2: Structurization. At Stage 2, raw dictionary entries were converted into structured metadata records by an LLM-based extraction pipeline. Given the scale of JieZi-Dataset (over 500K entries), exhaustive manual verification was impractical at this stage. Instead, to verify dataset quality, we randomly sampled 10% of the extracted metadata records; experts inspected each sample by comparing it directly with the source text. Errors were found in approximately 1% of the inspected subset. The most common errors were missing structured fields and cross-source confusion. For example, the pipeline might misattribute a semantic explanation originally associated with the Oracle Bone form to the Bronze Inscription field of the same character.

<Template> Extraction (extend)

```
# Warning
- Do not miss any numbered sense.
- Do not output any extra characters outside this JSON.

[System lines appended by code in _build_messages]
Again: output only one valid JSON object. Do not output code blocks. Do
not output explanatory text.
If fields are missing, fill them according to the prompt rules so the
schema is complete.

User message
- The user content is `entry_text`, i.e., preprocessed text from each txt
file, not a fixed template sentence.
- Preprocessing in `_preprocess_entry`:
  - normalize line breaks and Unicode;
  - remove the full [Glyph Form] section;
  - keep only: original char, pinyin, derived char, variant char, structure,
original meaning, evolution;
  - send the cleaned entry text as the user content.

2) Extra system prompt used in retry with larger max_tokens
Your previous output may have been truncated due to length limits,
causing incomplete JSON.
Please regenerate the same JSON completely, and it must be strictly
parseable JSON.
To reduce length, output compact JSON (no indentation, no extra spaces,
no line breaks), but do not omit any field/sense/example.

3) Extra system prompt used in paged fallback
You must use paged output mode.
Output only one valid JSON object, and you may include an extra field
`next_index` (integer).
Starting from index {start_index}, output at most {page_size}
`meaning_chunks` items in the original numbered order.
Each `meaning_chunks` item must include complete original examples in
`examples` (no omission).
If more senses remain, set `next_index` to the next sense index;
otherwise set `next_index=0`.
To reduce length, output compact JSON (no indentation, no extra spaces,
no line breaks).
```

Figure 5: Prompt of metadata construction(extend).

For JieZi-Bench, all structured metadata records were manually checked. Errors requiring correction were found in approximately 5% of entries, again higher than JieZi-Dataset due to the stricter acceptance criteria applied to the benchmark and the additional complexity of cross-source alignment.

Stage 3: VQA Generation and Verification. At Stage 3, VQA pairs were generated from the verified metadata by an LLM guided by expert-designed templates. The use of templates is essential because unconstrained LLM generation would suffer from three problems: (1) *format inconsistency*, where question phrasing and answer granularity vary unpredictably, hindering standardized evaluation; (2) *uneven subtask coverage*, where the LLM tends to favor subtasks it handles well while underrepresenting harder ones such as COME and EVOI; (3) *hallucination risk*, where the LLM may fabricate etymological explanations or evolutionary paths absent from the source dictionaries. Our template-guided approach addresses all three issues. Each template prescribes the question format and

<Template> VQA generation

Please output a structured JSON based on the image. Required fields: glyph type, character formation method, structure, special structure, components, original meaning, diachronic glyph evolution.

Please generate a standard answer that can be used directly for SFT from this single-character image.

Your output must satisfy all requirements below:
1. Output only one JSON object.
2. Use exactly these fields: glyph type, character formation method, structure, special structure, components, original meaning, diachronic glyph evolution.
3. The field "glyph type" must be consistent with the sample label "{glyph type}".
4. "structure" and "components" must follow the actual form in the current image stage.
5. "original meaning" and "diachronic glyph evolution" must be grounded in context, especially historical divergence, later forms, and simplification merges.
6. For simplified-form images, modern mainstream meaning is allowed for "original meaning", but the source chain must be explicit in "diachronic glyph evolution".
7. If weak hints conflict with the image, trust the image.
8. "original meaning" must align with the candidate list first; only minor paraphrase is allowed when strongly supported.
9. If "structure" is "single component", "special structure" must be a short description; otherwise it must be an empty string.
10. If the same component appears multiple times in different positions, split it into separate keys with position suffixes.

Recommended internal order:
- Identify the glyph stage.
- Identify the actual structure in this stage.
- Identify visible components and their functions.
- Determine the original meaning.
- Complete the diachronic evolution chain.

Current sample info:
- Sample ID: {sample_id}
- Image filename: {image_name}
- Weak glyph label hint: {glyph_type}

Failure correction hint:
{retry_guidance}

Auxiliary context (for reference, not for literal copying):
{labeling_context_json}

Please identify the modern dictionary headword shown in the image.

What is the dictionary headword of the character in this image?

Please answer the headword of this image directly.

Please classify the glyph type of the character in the image (e.g., Oracle Bone Script, Bronze Script, Seal Script, Clerical Script, Regular Script, Warring States Script, Simplified).

Which glyph type does this image belong to?

Please provide the glyph category of the character image.

What is the modern dictionary headword corresponding to this glyph image?

Please identify the modern dictionary headword of the character in the image.

Based on the image, what is the character headword in a modern dictionary?

Which glyph type is shown in the image?

...

Figure 6

expected answer scope for a given subtask, ensuring uniform coverage and consistent granularity. The LLM then generates natural-language VQA pairs grounded in the metadata fields specified by the template, combining the scalability of LLM generation with the controllability of structured constraints.

Despite these safeguards, template-guided generation introduces its own errors. The generation process can produce image-content mismatches (e.g., pairing a glyph image with an answer describing a different script period), and natural-language answers may deviate from the source text in phrasing or scope. Moreover, VQA-level verification covers broader dimensions than Stage 2, including not only field-level correctness but also the naturalness of question-answer formulations and the appropriateness of paired images, which increases the likelihood of flagging issues.

For JieZi-Dataset, we randomly inspected 5,000 VQA instances. Since the underlying metadata had already been verified at Stage 2, the sampling ratio was reduced accordingly. Errors were found in approximately 3% of the inspected subset; the slightly higher rate compared to Stage 2 reflects the new error sources and broader verification scope described above rather than upstream propagation.

For JieZi-Bench, all VQA pairs were manually checked, since benchmark reliability is critical for evaluation. Errors requiring correction were found in approximately 10% of entries, consistent with the pattern observed at earlier stages: the benchmark's stricter acceptance criteria and cross-source complexity lead to a higher correction rate.

Overall, JieZi-Dataset adopts random spot-checking at Stages 2 and 3 to balance scale and quality, while JieZi-Bench uses exhaustive manual verification at all stages. These combined strategies ensure that JieZi-Bench achieves expert-level reliability for evaluation, while JieZi-Dataset maintains sufficient quality for training at scale.

C Experimental Details and Full Results

C.1 Additional Training Details

In this section, we present the training details. The Parameter settings of instruction tuning are shown in Table 3. All experiments are completed on 8 Ascend 910B NPUs in 72 hours.

Table 3: Training settings and hyper-parameters for different model scales.

Setting	Qwen3.5-2B	Qwen3.5-4B	Qwen3.5-9B
Batch size	128	128	64
Learning rate	1e-5	1e-5	1e-5
LR schedule	cosine	cosine	cosine
Warmup ratio	0.05	0.05	0.05
Weight decay	0.1	0.1	0.1
Optimizer	AdamW	AdamW	AdamW
Max length	2048	2048	1024
Epochs	5	5	5
Precision	bf16	bf16	bf16
Hardware	8 910B	8 910B	8 910B
Training time	96 NPU hours	192 NPU hours	384 NPU hours

C.2 Ablation Studies

Tab. 4 demonstrates the impact of different data construction stages. We report results on CHAR, COMF, COMI and EVOI. Fine-tuning with the initial extracted and structured data (+ Structured metadata) provides a clear baseline boost, significantly improving metrics like COMF (from 13.7 to 29.7). Applying VQA Generation to format the training data further elevates the performance across the board. Ultimately, integrating open-source datasets to enhance visual diversity yields the best results. This consistent improvement validates the necessity and effectiveness of each individual part in our proposed pipeline.

Table 4: Ablation study on different data construction stages. Struct.: structured metadata extraction; VQA: VQA data generation; Open-src: open-source data augmentation.

Struct.	VQA	Open-src	CHAR	COMF	COMI	EVOI	
						FAC	SCE
			18.9	13.7	17.0	7.9	20.2
✓			27.0	29.7	24.1	15.4	26.3
✓	✓		35.3	31.4	37.1	25.6	31.9
✓	✓	✓	41.8	32.7	41.3	27.9	36.7

C.3 Human Validation of the LLM-as-a-Judge

To rigorously validate the reliability of our LLM-as-a-Judge protocol for the Evolution Interpretation (EVOI) task, we conducted a human-LLM agreement study. We randomly sampled 200 responses in total across 4 representative models to ensure a diverse distribution of response qualities. Three human experts with backgrounds in Chinese paleography independently evaluated these responses on a scale of 1 to 5, focusing on two dimensions: Fact Alignment (FAC) and Scholarly Expression (SCE).

We computed the Pearson (r) and Spearman (ρ) correlation coefficients between the automated LLM judge scores and the human ground truth.

As shown in Tab. 5, the LLM judge exhibits strong correlations (> 0.60) with expert evaluations across both metrics. The results confirm that the LLM judge aligns closely with human scholarly standards, demonstrating its effectiveness and reliability as an automated evaluation metric for ancient character exegesis.

Table 5: Correlation between human expert ratings and LLM judge scores on the EVOI task.

Metric	Pearson (r)	Spearman (ρ)
FAC	0.68	0.71
SCE	0.61	0.64

We provide the full prompts used in our LLM-based judging protocol for EVOI. Fig. 7 presents the prompt for evaluating **Fact Alignment**, while Fig. 8 presents the prompt for evaluating **Scholarly Expression**. We used Doubao-seed-2-0-lite-260215 [4] as our LLM-as-a-judge model.

<Template> LLM-judge FAC

You are a strict evaluator for ancient Chinese character tasks.
This time, you will evaluate only one dimension: fact alignment.
Your task is not to evaluate the entire response. You must only evaluate whether the sentences describing the "historical character shape evolution" in the prediction are factually consistent with the "historical character shape evolution" in the reference answer.
You must adhere to the following principles:
Base your judgments solely on the provided Ground Truth. The use of external knowledge is strictly prohibited.
Do not evaluate writing style, wording, or eloquence.
Do not award high scores simply because the prediction "sounds like" a professional ancient character analysis.
Differences in phrasing from the reference answer are acceptable, but altering factual information is not.
ground_truth_full serves only as auxiliary background context. The primary targets for scoring
remain ground_truth_evolution and prediction_evolution.
Partial credit is allowed for this dimension: if the prediction has substantial points of alignment regarding key structures, component relationships, historical periods, or evolutionary directions, award corresponding points. Do not automatically penalize the score to 0 or 1 merely because some errors exist.
Please focus your review on the following:
Alignment of the core structural composition (核心构形).
Alignment of relationships between key components (关键部件关系).
Alignment of key historical periods/stages (关键时代阶段).
Alignment of the evolutionary direction (演变方向).
Whether the character has been conflated with a different character or an entirely different structural system.

Scoring Rubric (Must be exactly 0, 1, 2, 3, or 4):
4: Highly consistent with the reference answer; almost zero factual errors.
3: The main narrative is largely consistent, but there are some factual deviations or inaccuracies in minor details.
2: Contains verifiable and clear points of alignment (e.g., correctly identifies at least *some* structures/components/periods/directions), but concurrently contains obvious factual errors. Overall reliability is moderate.
1: Shows only minimal, superficial similarity; the vast majority of key facts are incorrect.
0: Fundamentally contradicts the reference, is completely blank, or describes the shape evolution of an entirely different character/system.

Mandatory Calibration Rules:
If the prediction is blank: score = 0
If the core structural composition, core components, or main evolutionary direction are vastly incorrect: generally score <= 2
If the character is misidentified as a different structural system, or the main factual narrative is fundamentally misplaced: score = 0
If the response relies purely on generic clichés/filler with almost no verifiable facts: the score cannot exceed 2
If the prediction contains multiple key points that align with the reference (even if accompanied by errors), the score generally should not be penalized to 0 or 1.
Do not default to grading most samples as 3 or 4 (avoid score inflation).

Now, please evaluate the following sample to be rated according to the scoring rules above.
Ground Truth: {}
Output of model: {}

Figure 7: Prompt of the llm-judge of Fact Allignment.

C.4 Main Results on JieZi-Bench and Analysis

Due to space limitations in the main text, we present the complete experimental results in Table 6, including additional models not shown in the main paper. Below, we provide a detailed analysis of the key findings.

Table 6: Main results on JieZi-Bench across four paleographic levels: L1 (Basic Info: CHAR, SCRC), L2 (Glyph Form: STRC, COMR, COMF, COMI, FORC), L3 (Meaning: ORIM), and L4 (Evolution: COME, EVOI-FAC/SCE). All metrics are scaled to 0-100.

Method	L1		L2					L3	L4		
										EVOI	
	CHAR↑	SCRC↑	STRC↑	COMR↑	COMF↑	COMI↑	FORC↑	ORIM↑	COME↑	FAC↑	SCE↑
Closed-source MLLMs											
GPT-5.4 Thinking [17]	10.4	39.2	50.4	13.8	11.9	10.3	33.1	59.9	12.4	10.2	16.8
Gemini 3.1 Pro [10]	29.9	76.6	65.7	31.9	29.9	24.8	56.1	62.8	30.3	22.3	27.1
Gemini 3.1 Flash [9]	25.7	56.5	53.5	24.6	22.5	19.1	44.8	54.1	23.2	17.5	23.1
Claude opus 4.6 [1]	22.9	65.8	63.5	28.0	26.3	22.3	50.6	59.3	26.7	19.8	26.3
Doubao-Seed-2.0-pro [3]	36.6	68.5	69.8	37.8	34.4	29.9	59.9	63.2	36.1	31.7	38.7
Open-source MLLMs											
Kimi-K2.5 [21] (170B)	31.5	70.1	68.8	39.2	36.6	31.5	58.1	66.6	36.4	28.3	34.6
GLM-4.6V [22] (107B)	18.2	49.7	48.9	21.8	20.4	18.2	42.2	48.2	20.9	15.5	20.2
Qwen3.5-397b-a17b [18]	26.2	67.1	69.4	32.1	29.9	26.5	54.4	59.5	28.3	22.2	30.7
Qwen3.5-35b-a3b [18]	26.6	61.4	66.5	32.4	29.3	27.3	57.8	65.2	31.0	19.6	25.6
InternVL3.5-8B [23]	00.1	09.0	14.6	0.0	0.0	0.0	21.4	58.0	0.0	24.8	0.1
InternVL3.5-4B [23]	0.0	0.0	0.9	0.0	0.0	0.0	8.2	24.3	0.0	0.0	0.0
tonggu-vl-2B [5]	3.1	05.9	0.2	0.0	0.0	0.0	4.6	12.9	0.0	1.1	0.2
Qwen3.5-2B [18]	18.9	22.6	31.6	17.3	13.7	17.0	30.7	62.3	12.9	7.9	20.2
Qwen3.5-2B + JieZi-Dataset	41.8+22.9	61.7+39.1	72.7+41.1	42.6+25.3	32.7+19.0	41.3+24.3	63.3+32.6	65.7+3.4	28.3+15.4	27.9+20.0	36.7+16.5
Qwen3.5-4B [18]	3.6	6.6	1.0	4.5	3.9	4.3	5.9	6.5	3.6	2.0	3.6
Qwen3.5-4B + JieZi-Dataset	49.0+45.4	67.6+61.0	74.1+73.1	47.2+42.7	37.9+34.0	42.4+38.1	64.8+58.9	65.9+59.4	33.2+29.6	28.5+26.5	37.2+33.6
Qwen3.5-9B [18]	22.3	51.0	40.2	27.4	24.6	26.1	49.1	64.7	22.4	15.0	23.5
Qwen3.5-9B + JieZi-Dataset	**52.6**+30.3	**80.1**+29.1	**74.7**+34.5	**48.1**+20.7	**38.4**+13.8	**45.7**+19.6	**66.5**+17.4	**66.7**+2.0	**45.4**+23.0	**32.5**+17.5	**40.6**+17.1

Scaling law holds within model families. The supplementary table includes additional model variants not reported in the main paper, enabling within-family comparisons among general MLLMs. Across all families, larger variants consistently outperform smaller ones. Gemini 3.1 Pro surpasses Gemini 3.1 Flash on every metric (e.g., CHAR: 29.9 vs. 25.7; SCRC: 76.6 vs. 56.5), Qwen3.5-397b-a17b outperforms 35b-a3b (e.g., SCRC: 67.1 vs. 61.4), and InternVL3.5-8B consistently exceeds its 4B counterpart (e.g., ORIM: 58.0 vs. 24.3). These within-family comparisons, together with the cross-family results in the main paper, confirm that the general scaling trend holds for ACCE.

Domain-specialized pretraining does not guarantee ACCE proficiency. tonggu-vl-2B [5] is a model specifically fine-tuned on massive ancient Chinese document corpora for tasks such as text recognition and reading comprehension. Despite this domain-specialized pretraining, it scores below 5% on most ACCE subtasks (e.g., CHAR: 3.1, STRC: 0.2, COMR/COMF/COMI: 0.0) and achieves only 12.9 on ORIM. This reveals a critical distinction: *recognizing and reading ancient texts* is fundamentally different from *performing scholarly exegesis*, which requires structured reasoning about glyph form, component function, semantic origin, and diachronic evolution. The failure of tonggu-vl demonstrates that ACCE poses genuinely novel challenges beyond the reach of existing domain-specific models, thereby validating both the task formulation and the necessity of JieZi-Dataset as a dedicated training resource. Conversely, once equipped with JieZi-Dataset, even a 9B model outperforms general MLLMs that are one to two orders of magnitude larger (e.g., Kimi-K2.5 170B, GLM-4.6V 107B, Qwen3.5-397b) on nearly every subtask, further confirming that high-quality domain-specific data is a far more effective lever than model scale alone for knowledge-intensive tasks like ACCE.

Remark on the anomalous behavior of Qwen3.5-4B. As shown in Tab. 6, few-shot Qwen3.5-4B scores near zero across all eleven ACCE subtasks (e.g., CHAR: 3.6, STRC: 1.0, ORIM: 6.5), performing substantially below both the smaller 2B and the larger 9B variants. We verified that all three models share identical inference pipelines, prompts, and decoding parameters, ruling out implementation errors. Manual inspection of the 4B outputs (Fig. 9) reveals that the model tends to reproduce content from the few-shot demonstrations rather than generating answers grounded in the current query, producing outputs that bear no meaningful correspondence to the target glyph.

Since Qwen3.5's training details are not publicly disclosed, the root cause remains unclear. We identify two plausible contributing factors. (1) *Copying bias in few-shot ICL*: the model reproduces demonstration content instead of reasoning about the current query. This behavior directly aligns with prior findings that LLMs can develop a copying bias under in-context learning, reproducing surface patterns from examples rather than inducing the underlying

<Template> LLM-judge SCE

You are a strict evaluator for paleography tasks, evaluating only the scholarly expression dimension.
Evaluate **only** the "historical character shape evolution" sentences in the prediction. Judge if the expression is professional, rigorous, and matches a paleographic dictionary (lexicographical) style, rather than reading like layman's explanations or empty clichés.
Core Principles:
GT Only: Base judgments strictly on the provided Ground Truth. No external knowledge.
Style over Fact (Mostly): Evaluate terminology, logic, and professional tone. However, professional tone wrapped around a completely wrong evolutionary narrative cannot score high.
Quality > Quantity: Do not reward length unless it provides valid information gain.
Dictionary Style: Prefer concise, objective summaries of structure and evolution. Penalize lyrical, heuristic, or layman-oriented tones.
Scoring Rubric (Strictly 0, 1, 2, 3, or 4):
4: Highly mature, dictionary-level expression. Precise terminology, restrained judgments, rigorous logic, and substantial content with zero fluff.
3: Professional and solid. Clear lexicographical traits and good logic, but contains minor looseness or slight layman phrasing.
2: Readable but lacks true professional depth. Unstable style, leans toward generic explanatory text, or contains obvious verbosity/clichés.
1: Loose and verbose. Strong layman tone, high fluff, lacks rigor, or merely dresses up generic content with scattered jargon.
0: Blank, or entirely empty clichés/slogans with no valid professional information.
Mandatory Calibration Rules:
If blank: score = 0
If longer than GT but mostly adds repetitive fluff/no new info: score <= 2
If using generic layman phrases (e.g., it can be seen, shows the wisdom of the ancients): score <= 2
If built on obviously wrong factual narratives or incorrect structural judgments: score <= 2
If terminology is chaotic or reads like a generic essay: score <= 1
If merely using jargon like clericalization to mask empty/wrong content: score <= 1
Be Strict: 4 is for exceptionally mature, dictionary-quality text only. Do not default to 3. Do not give high scores just because the text is long or reads smoothly.

Now, please evaluate the following sample to be rated according to the scoring rules above.

Ground Truth: {}

Output of model: {}

Figure 8: Prompt of the llm-judge of Scholarly Expression.

task [15, 16]. (2) *U-shaped scaling*: the 2B (functional) → 4B (collapsed) → 9B (functional) pattern is consistent with the U-shaped scaling phenomenon, where medium-sized models are drawn toward easier competing behaviors instead of the target task [14, 24].

After fine-tuning on JieZi-Dataset, Qwen3.5-4B fully recovers to the expected scaling order, achieving the largest absolute improvements among the three model scales (e.g., STRC: +73.1, ORIM: +59.4 vs. 2B's +41.1/+3.4 and 9B's +34.5/+2.0). This confirms that the bottleneck is not model capacity but misaligned few-shot behavior, which domain-specific supervision effectively overrides [27].

References

[1] Anthropic. 2026. Claude Opus 4.6 System Card. https://www-cdn.anthropic.com/14e4fb01875d2a69f646fa5e574dea2b1c0ff7b5.pdf
[2] Françoise Bottéro. 1996. Review of The Origin and Early Development of the Chinese Writing System, by William G. Boltz. *Journal of the American Oriental Society* 116, 3 (1996), 574–577. https://doi.org/10.2307/605196
[3] ByteDance Seed. 2026. Seed2.0 Model Card. https://seed.bytedance.com/seed2
[4] ByteDance Seed Team. 2026. Seed 2.0 Official Launch. Official blog post. https://seed.bytedance.com/en/blog/seed2-0-%E6%AD%A3%E5%BC%8F%E5%8F%91%E5%B8%83 Accessed: 2026-04-09.
[5] Jiahuan Cao, Yang Liu, Peirong Zhang, Yongxin Shi, Kai Ding, and Lianwen Jin. 2025. TongGu-VL: Advancing Visual-Language Understanding in Chinese Classical Studies through Parameter Sensitivity-Guided Instruction Tuning. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 11111–11120.
[6] Diego Chapinal-Heras and Carlos Díaz-Sánchez. 2023. A review of AI applications in Human Sciences research. *Digital Applications in Archaeology and Cultural Heritage* 30 (2023), e00288.
[7] Yang Chi, Fausto Giunchiglia, Chuntao Li, and Hao Xu. 2024. Ancient Chinese Glyph Identification Powered by Radical Semantics. In *Findings of the Association for Computational Linguistics: ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 12065–12074. https://doi.org/10.18653/v1/2024.findings-acl.718
[8] Wiebke Denecke, Wai-yee Li, and Xiaofei Tian. 2017. *The Oxford handbook of classical Chinese literature (1000 BCE-900 CE)*. Oxford University Press.
[9] Google DeepMind. 2026. Gemini 3.1 Flash-Lite Model Card. https://deepmind.google/models/model-cards/gemini-3-1-flash-lite
[10] Google DeepMind. 2026. Gemini 3.1 Pro Model Card. https://deepmind.google/models/model-cards/gemini-3-1-pro
[11] David N Keightley. 1996. Art, ancestors, and the origins of writing in China. *Representations* 56 (1996), 68–95.
[12] Guolong Lai. 2019. On [Can] and [Xie]: Two Different Approaches to the Interpretation of Ancient Chinese Characters, Form-Oriented and Integrated Phonology-Form-Semantics. *Bulletin of the Jao Tsung-I Academy of Sinology* 6, 1 (2019), 187–224.
[13] Jing Li, Xueke Chi, Qiufeng Wang, Dahan Wang, Kaizhu Huang, Yongge Liu, and Cheng-Lin Liu. 2024. A comprehensive survey of oracle character recognition:

Prompt:

Role
Expert in Chinese paleography, Shuowen Jiezi and etymology…

Task

Output a strict JSON component analysis of the ancient character in the image, detailing each component's function, explanation, and evolution type …

Rules:

1. Rely SOLELY on the image; no external knowledge.
2. Never hallucinate unsupported details.
3. Output exactly ONE raw JSON object (NO markdown, NO comments, NO extra text).
4. Prioritize valid fields, aligned content, and conservative analysis. …

Required JSON Format
构件: Object of 1-4 key components.
Keys: Single characters/radicals (use brief desc ONLY if unnameable).
Values: Objects containing exactly:
功能: String or Array of: 表意, 表音, 符号.
演变类型: Choose ONE: 保留, 讹变, 简化, 区别, 繁化.
解释: Brief explanation …

Few-shot1: …
Few-shot2:

```
"构件": {
    "构件名": {
      "功能": "表意/表音/符号",
      "演变类型": "保留/讹变/简化/区别/繁化",
      "解释": "简短解释"
      }
  }
```

Qwen3.5-4B:

Just Copy !

```
"构件": {
    "构件名": {
      "功能": "表意/表音/符号",
      "演变类型": "保留/讹变/简化/区别/繁化",
      "解释": "简短解释"
      }
  }
```

Figure 9: A example of the output of Qwen3.5-4B

challenges, benchmarks, and beyond. arXiv:2411.11354 [cs.CV] https://arxiv.org/abs/2411.11354

[14] Ian R. McKenzie, Alexander Lyzhov, Michael Martin Pieler, Alicia Parrish, Aaron Mueller, Ameya Prabhu, Euan McLean, Xudong Shen, Joe Cavanagh, Andrew George Gritsevskiy, Derik Kauffman, Aaron T. Kirtland, Zhengping Zhou, Yuhui Zhang, Sicong Huang, Daniel Wurgaft, Max Weiss, Alexis Ross, Gabriel Recchia, Alisa Liu, Jiacheng Liu, Tom Tseng, Tomasz Korbak, Najoung Kim, Samuel R. Bowman, and Ethan Perez. 2023. Inverse Scaling: When Bigger Isn't Better. *Transactions on Machine Learning Research* (2023).

[15] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2022. Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*.

[16] Catherine Olsson et al. 2022. In-context Learning and Induction Heads. Transformer Circuits.

[17] OpenAI. 2026. GPT-5.4 Thinking System Card. https://deploymentsafety.openai.com/gpt-5-4-thinking/gpt-5-4-thinking.pdf

[18] Qwen Team. 2026. Qwen3.5: Towards Native Multimodal Agents. https://qwen.ai/blog?id=qwen3.5

[19] Zhongwei Shen. 2020. *A phonological history of Chinese*. Cambridge University Press.

[20] Adam D Smith. 2017. Early Chinese manuscript writings for the name of the Sage Emperor Shun, and the legacy of Warring States-period orthographic variation in early Chinese received texts. *Early China* 40 (2017), 63–88.

[21] Kimi Team, Tongtong Bai, Yifan Bai, Yiping Bao, S. H. Cai, Yuan Cao, Y. Charles, H. S. Che, Cheng Chen, et al. 2026. Kimi K2.5: Visual Agentic Intelligence. arXiv:2602.02276 [cs.CL] https://arxiv.org/abs/2602.02276

[22] V Team, Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale Cheng, Ji Qi, Junhui Ji, et al. 2025. GLM-4.5V and GLM-4.1V-Thinking: Towards Versatile Multimodal Reasoning with Scalable Reinforcement Learning. arXiv:2507.01006 [cs.CV] https://arxiv.org/abs/2507.01006

[23] Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. 2025. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency. *arXiv preprint arXiv:2508.18265* (2025).

[24] Jason Wei, Najoung Kim, Yi Tay, and Quoc V Le. 2023. Inverse Scaling Can Become U-Shaped. In *EMNLP*.

[25] Qiu Xigui. 1985. On the Methods of Studying Ancient Chinese Script. *Early China* 11 (1985), 301–316.

[26] Wen Xing. 2011. Paleographic, Historical, and Intellectual History Approaches to Warring States Manuscripts Written on Bamboo Slips: A Review Article. *Early China* 33 (2011), 233–262. https://doi.org/10.1017/S0362502800000298

[27] Biao Zhang, Zhongtao Liu, Colin Cherry, and Orhan Firat. 2024. When Scaling Meets LLM Finetuning: The Effect of Data, Model and Finetuning Method. In *ICLR*.